

Verifier-Assisted versus One-Shot LLM Intent→Configuration: A Paired NetOpsTraceBench Study

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Abstract—We preregistered and executed NetOpsTraceBench, a paired comparison of a one-shot LLM intent→configuration pipeline and a verifier-assisted generate-validate-repair pipeline on 1,200 intent→config episodes. Using gpt-5-mini, a deterministic NetOps validator, and Mininet-style emulation, we applied a preregistered binary episode success rule (validator accepts final config AND emulator shows no availability degradation above tolerance AND no automatic rollback). Of 1,200 planned paired tasks we completed 1,199 pairs. Baseline success = 1,196/1,199 (0.9974979149); guarded success = 1,199/1,199 (1.0000000000). The paired-success-rate difference (guarded - baseline) = 0.0025020851 (95% bootstrap percentile CI [0.0000, 0.0058381985], bootstrap_seed=1729, resamples=10,000); exact McNemar two-sided $p = 0.25$. The preregistered power target sought a 5 percentage-point minimum detectable effect; given the observed near-ceiling baseline, the study lacked power to detect much smaller absolute gains. Representative validator traces and provider audit logs show repair was invoked only three times. Under these controlled emulation and task-corpus conditions we find no statistically significant advantage for the verifier-assisted pipeline; we report artifacts, analytic deviation (Wilcoxon → exact McNemar for binary outcomes), and detailed execution accounting to support reproducible interpretation.

I. INTRODUCTION

Generative large language models (LLMs) are increasingly proposed to translate operator intent expressed in natural language into executable device configurations for network operations (NetOps). Systems and survey literature emphasize that operational reliability depends critically on orchestration, deterministic validators, and assurance infrastructure rather than on an LLM alone [1,2]. Generate-validate-repair (verifier-assisted) architectures are a canonical mitigation strategy: an LLM proposes candidate outputs, an automated verifier checks constraints and safety, and an automated repair loop iterates until the candidate satisfies verifier rules [3–5]. Despite conceptual support, publicly archived, preregistered, paired evidence measuring whether verifier assistance yields an operationally meaningful improvement (rollback-aware binary success) relative to one-shot generation is sparse.

We address this gap with NetOpsTraceBench: a preregistered, paired, emulator-grounded study comparing (A) baseline one-shot NL→config generation and (B) a guarded generate-validate-repair pipeline using the same shared initial candidate. The study asked: Can a reproducible, workflow-centered benchmark of paired intent→config episodes produce an objective paired binary success metric that reliably distinguishes verifier-assisted pipelines from one-shot generation across two

simulated topologies? Our contributions are (1) a preregistered experimental design and full execution artifacts (study ID rX4f7-1); (2) a paired analysis on 1,199 completed pairs using gpt-5-mini and a deterministic validator; (3) transparent reporting of an analytic deviation (final confirmatory test used an exact McNemar test appropriate for paired binary outcomes) and execution accounting that explains why the guarded pipeline rarely required repair.

II. RELATED WORK

Recent surveys and system papers position LLMs within AIOps/NetOps pipelines and emphasize the importance of verifiers, provenance, and end-to-end workflow evaluation [1,6]. Work on verifier-guided NL→formal translations and intent→topology compilation has shown the feasibility of generate-validate-repair approaches in domains such as security topology compilation and CTL/specification generation [3,7]. Empirical and conceptual studies document LLM failure modes relevant to NetOps—hallucination, incorrect semantics/units, overconfidence, provenance loss, and prompt injection—which motivate guarded architectures and run-time auditing [8–10]. Several recent prototypes demonstrate LLM-assisted configuration generation and misconfiguration detection but typically evaluate on constructed or small held-out sets [4,5,11]. That literature motivates workflow-centered evaluation metrics (trace quality, rollback measurement, canaries) rather than one-shot QA metrics; NetOpsTraceBench operationalizes a paired, rollback-aware binary success metric in a preregistered framework and archives the run artifacts for reproducibility.

III. METHODOLOGY

Overview and preregistration: The study was preregistered as NetOpsTraceBench (study_id = rX4f7-1) and the preregistration manifest (SHA-256 = 77abe8b493bde44484e0b3d02cfd7f9ec1621ff7e7847889082ae177a7ed5712) froze the experimental plan before any model calls. The preregistration enumerated confirmatory hypotheses (H1–H3), inclusion/exclusion rules, contamination thresholds, seeds, stopping rules, analysis executor (paired_binary_analysis_v1), primary estimand name (paired_success_rate_difference_guarded_minus_baseline), and the prespecified power target (MDE = 5 percentage

points; $\alpha=0.05$; power 0.8) given assumed baseline ≈ 0.70 and discordant-pair assumptions recorded in the plan.

Task corpus and topology clusters: The benchmark scope defined 1,200 distinct operator-intent tasks split into two simulated topology clusters: edge (600) and core (600). Each task is generated deterministically by `deterministic_netops_task_generator_v5` (`generator_version = 5.0`) using frozen per-task `generation_seed` values. Task payloads contain structured fields: `initial_state`, `expected_final_state`, `explicit_required_sequence` arrays, `allowed_touched_fields`, `workflow_policies`, and a declared difficulty level (e.g., "medium", "hard", "very_hard"). Tasks are labeled synthetic or consented and carry deterministic provenance metadata used by contamination checks. Task and schema versions are 1.0.

Contamination and inclusion/exclusion: The preregistration specified conservative contamination detection: token/subsequence overlap threshold $T_{\text{contam}} = 0.85$ (normalized overlap) and optional embedding-similarity check (cosine threshold 0.92) for flagged items. Pairs were excluded automatically if contamination checks exceed thresholds, if both episodes fail to produce parsable vendor-rendered configs after one automated re-execution attempt, or if emulator instrumentation deterministically fails to measure availability. Exclusions are deterministic, logged, and reported; no human adjudication occurs. The missingness plan defines primary analysis as `complete_pair_primary` (exclude flagged pairs) and a sensitivity analysis that treats excluded pairs as failures (`complete_pair_primary_with_failure_as_zero_sensitivity`).

Pipeline implementations and orchestration semantics: We implemented two paired conditions per task using a shared initial candidate generation approach (`generation_semantics = "shared_initial_candidate"`). Adapter orchestration used `hosted_netops_gvr_v1`. Model requests used the `requested_model = gpt-5-mini` via provider recorded as `openai_responses`; `execution/model_configuration.json` records `requested_model = gpt-5-mini`, `max_output_tokens = 1500`, `maximum_attempts_per_call = 1`. Per preregistration, `initial_generation_calls_per_task = 1` and `maximum_repair_calls_per_task = 1`; `retrieval_augmented_generation = false`; `independent_condition_generation = false` (guarded and baseline share the same initial candidate in each pair to isolate verifier and repair effects).

Baseline (one-shot) condition: For each task the LLM receives the frozen, schema-specified intent and returns a single candidate configuration (the `shared_initial_candidate`). The baseline episode records that candidate and proceeds to deterministic parsing and emulator preflight checks without invoking the automated repair loop.

Guarded (verifier-assisted) condition: The guarded pipeline feeds the same `shared_initial_candidate` to `deterministic_netops_validator_v5` (`validator_version = 5.0`). If the validator accepts the candidate, the episode proceeds to emulator execution; if the validator rejects the candidate the pipeline may invoke an automated repair iteration (a single additional LLM call per preregistered `maximum_repair_calls_per_task =`

1) that produces a repaired candidate which is then revalidated. Repair invocation is conditional and limited by the frozen retry policy; all repair prompts, provider traces, and repair results are archived. The scoring rule used by the validator is recorded as `"full_intent_constraint_validation"` and `scoring_status` entries are emitted (COMPLETED or error codes) into per-episode scoring artifacts.

Deterministic validator and emulator checks: `deterministic_netops_validator_v5` performs canonical parsing, counts `parsed_command_count` and `required_command_count`, enforces `workflow_policies` (e.g., `must_be_down_for_fields`, `minimum_active_in_groups`), and produces a `normalized_configuration` string. The preregistered binary episode success requires (1) validator acceptance of the final configuration (`validation_after.valid == true`), (2) emulator execution showing no availability degradation above the preregistered tolerance, and (3) no automatic rollback event during simulated execution. Validator traces include structured diagnostics: `observed_touched_field_count`, `violation_codes`, difficulty metadata (`changed_interface_count`, `dependency_rule_count`, `pattern`), and `repair_applied` boolean. These diagnostics were used for exploratory failure-mode catalogs.

Analysis plan and confirmatory procedure: The preregistration specified a paired analysis using `paired_binary_analysis_v1`. The primary estimand is the `paired_success_rate_difference_guarded_minus_baseline`. The preregistration anticipated using a nonparametric paired test and bootstrap CI computation. Because observed outcomes are binary, the archived confirmatory analysis executed an exact McNemar two-sided test for paired binary outcomes and computed a paired bootstrap percentile 95% CI using frozen `bootstrap_seed = 1729` and `bootstrap_resamples = 10,000`; both the contingency table and bootstrap procedure are archived. Missingness, exclusions, and deviations are reported deterministically; no post-hoc inclusion decisions were made.

Stopping rules, provider audit, and execution accountability: The run obeyed the preregistered stopping rule (halt if model calls reach `maximum_model_calls = 2,400`). Execution produced `provider_call_audit` entries: `hosted_model_call_event_count = 1,203`; `completed_hosted_model_call_event_count = 1,202`; `failed_hosted_model_call_event_count = 1`; `stage_counts`: `shared_initial_generation = 1,200` and `repair = 3`. Execution manifest and `deterministic_reconciliation` artifacts track `completed_episode_count = 2,398`, `failed_episode_count = 2`, and `complete_pair_count = 1,199`. All `raw_model_outputs`, provider traces, task manifests, transformation manifests, validator traces, scoring artifacts, and analysis logs (including `analysis/paired_contingency_table.csv`, `analysis/condition_summary.csv`, `analysis/missingness_summary.csv`, and `analysis/paired_difference_figure.svg`) are archived to permit exact replay of the analysis executor and CI computation.

Power rationale and sensitivity: The preregistered power

plan targeted an absolute paired increase of 5 pp favoring guarded (MDE = 0.05) under assumed baseline ≈ 0.70 and discordant-pair assumptions; those assumptions and sensitivity grids (p0 in [0.5,0.8], discordant rates in [0.1,0.4]) are stored in the preregistration. The manifest records that the observed baseline was near ceiling (≈ 0.9975), which reduces power to detect small absolute improvements; the archived bootstrap CI and discordant counts provide the quantitative basis for that interpretation.

Reproducibility and artifact governance: All seeds, code for task generation, validator rules, repair prompts, provider traces, and analysis code are published in the run artifacts. The run produced `artifact_count` = 8,408 and a full hash manifest; representative per-episode scoring JSONs (schema version 1.0) show the `normalized_response`, `validator_trace` entries, and `score_reason_code` values used to compute the binary success outcomes. The methodology deliberately prioritizes deterministic components and explicit logging to ensure that the paired comparison isolates the verifier/repair mechanism while keeping generation variability constrained by `shared_initial_candidate` semantics.

IV. RESULTS

Execution and accounting (archived): Planned `episode_count` = 2,400 (1,200 pairs); `completed_episode_count` = 2,398; `failed_episode_count` = 2; `completed_pairs` = 1,199. Provider audit: `hosted_model_call_event_count` = 1,203; `completed_hosted_model_call_event_count` = 1,202; `failed_hosted_model_call_event_count` = 1. Stage counts show `shared_initial_generation` = 1,200 and `repair` = 3 (i.e., the guarded pipeline triggered repair calls only three times). `Artifact_count` = 8,408 and the run completed at 2026-08-30T03:15:12.141890+00:00; representative artifact examples and validator traces are archived (e.g., `execution/scoring/task-000001-baseline.json`).

Primary confirmatory outcomes (archived analysis): `complete_pair_count` = 1,199; `baseline_success_count` = 1,196 (`baseline_success_rate` = 0.9974979149291076); `guarded_success_count` = 1,199 (`guarded_success_rate` = 1.0). Paired contingency: `n11` = 1,196; `n10` = 3 (`guarded=1 baseline=0`); `n01` = 0; `n00` = 0 (see `analysis/paired_contingency_table.csv`). Paired-success-rate difference (`guarded` - `baseline`) = 0.002502085070892446. Exact McNemar two-sided p = 0.25. Paired bootstrap percentile 95% CI (`bootstrap_seed` = 1729; 10,000 resamples) = [0.0000, 0.005838198498748957]. Missingness summary: `failed_baseline_episodes` = 1; `failed_guarded_episodes` = 1; `both_missing_pairs` = 1 (see `analysis/missingness_summary.csv`).

Representative diagnostics: Validator traces in scoring artifacts show that, for typical tasks, the initial shared candidate satisfied intent constraints and validator checks (e.g., `execution/scoring/task-000001-baseline.json` shows `validation_after.valid` = true and `repair_applied` = false). The low

repair invocation count (3) is corroborated by the provider audit stage_counts. These artifact-grounded diagnostics indicate that under the chosen validator configuration and synthetic task corpus the LLM initial candidates were accepted in the vast majority of episodes, leaving little headroom for guard-triggered improvement.

V. DISCUSSION

Statistical and operational interpretation: The exact McNemar test (p = 0.25) and the paired bootstrap CI [0.0000, 0.0058381985] both indicate the observed guarded advantage is not distinguishable from zero under our confirmatory criterion. The empirical discordant counts (`n10` = 3, `n01` = 0) directly determine the paired estimate: `paired_difference` = $3/1,199 = 0.0025020851$. Because the preregistered MDE was 0.05 (5 percentage points), the study—while sized for that target—was not powered to detect the much smaller absolute difference we observed. Concretely, observing a 5 pp absolute paired difference on the realized `complete_pair_count` (1,199) would require approximately 60 discordant pairs ($0.05 \times 1,199 \approx 59.95$), but we observed only three discordants; this arithmetic shows why the planned MDE and realized discordant count are inconsistent with detecting the tiny observed effect.

Artifact-grounded diagnostics explain why repairs were rare. The provider audit and stage counts record only three repair invocations (`repair` = 3) and `model_calls_used` = 1,203, corroborated by the validator traces where `validation_after.valid` = true and `repair_applied` = false for representative tasks (see `execution/scoring/task-000001-baseline.json`). Two interacting design choices reduced headroom for verifier benefit: (1) the synthetic task generator produced well-specified intents with canonical `required_sequences` (`deterministic_netops_task_generator_v5`), and (2) the deterministic validator (`deterministic_netops_validator_v5`) used parsing and constraint checks that accepted many reasonable initial candidates. These two factors are visible in representative task_manifest entries (`required_sequence` and `difficulty` fields) and in the scoring artifacts where `parsed_command_count` == `required_command_count` and `violation_count` == 0 for typical episodes. Together they explain the low frequency of repair-triggered iterations and the near-ceiling baseline success ($\approx 99.75\%$).

Reproducibility and forensic trace evidence: All numerical claims above are reproducible from archived artifacts. The confirmatory analysis used `bootstrap_seed` = 1729 with 10,000 resamples to compute the percentile CI (`analysis/analysis_log.jsonl`), and the paired contingency table is published (`analysis/paired_contingency_table.csv`). Execution accounting (`execution/execution_manifest.json`) reports `started_at_utc` and `completed_at_utc`, `model_calls_used`, `completed_pairs`, and `failed_episodes`. `Artifact_count` = 8,408 and representative scoring JSON entries include `validator_trace` objects showing `validation_before/after`, `observed_touched_field_count`, and `repair_applied` flags. Those deterministic traces permit exact reproduction of the numeric summaries without human adjudication.

Threats to validity made concrete by artifacts: Internal validity is high because task generation, seeds, and deterministic validators were frozen in the preregistration (preregistration SHA-256 = 77abe8b4...5712) and automation enforced exclusion rules; the `missingness_summary.csv` documents the two failed episodes and the single `both_missing_pair`. Construct validity depends on how well the binary episode success rule (validator acceptance + emulator availability tolerance + no rollback) maps to operator-valued correctness; the validator’s acceptance thresholds and the emulator’s availability tolerance are parameters embodied in the validator traces and `transformation_manifest` and therefore materially affect measured outcomes. External validity is limited: only one model family (requested_model = gpt-5-mini) was invoked (execution/model_configuration.json), tasks were synthetic/consented and vendor-agnostic, and emulation is Mininet-style rather than vendor-specific hardware; these constraints appear in task_manifest entries and in the emulator preflight logs. Conclusion validity is constrained by the small number of discordant pairs: standard paired tests rely on sufficient discordance to support inference, and here the discordant mass (3/1,199) is too small to yield narrow hypothesis tests for small absolute differences.

Implications for benchmark design and operational practice: The artifacts show that when initial generation already satisfies deterministic verifier rules for well-specified synthetic tasks, guarded architectures will seldom be exercised, and measured binary success gains will be tiny. To reveal verifier value in future studies, designers should (a) increase nominal task ambiguity or adversarial content so that initial candidates more frequently violate intent constraints, (b) vary validator strictness to control repair invocation rates, and (c) test multiple model families or fine-tuned variants to measure model-validator interactions. The archived run provides concrete knobs: change task_manifest difficulty fields, adjust validator violation thresholds, or expand maximum_repair_calls_per_task to study repair dynamics—each change is reproducible because seeds, manifests, and validator traces are preserved.

Operational diagnostics and sensitivity: The low repair count (3) correlated with specific task patterns (deterministic task difficulty levels labelled "medium" or "hard" in representative manifests) but was insufficient to permit reliable subgroup inference. Exploratory artifacts (analysis/condition_summary.csv and analysis/missingness_summary.csv) can be used to stratify by difficulty and re-estimate discordant rates; however, such exploratory subgroup tests remain underpowered without increasing task diversity or sample size. The run’s provider audit also shows a single failed hosted_model_call_event_count = 1; recovery and retry rules are logged and the stopping_rule was enforced without manual intervention.

Concluding operational guidance: The null confirmatory result—guarded advantage not statistically significant—should be interpreted as conditional on the task corpus, validator con-

figuration, and requested model. The run artifacts consistently point to limited headroom (near-ceiling baseline) and low repair invocation as the proximate causes. For practitioners, the key takeaway is not that verifiers are universally unnecessary, but rather that their measured value depends strongly on task ambiguity and validator policy. We provide the full artifact set (task manifests, validator traces, provider audits, analysis logs, and seeds) to enable sensitivity experiments that vary these parameters and thereby expose conditions under which verifier scaffolding yields operationally meaningful improvements.

VI. LIMITATIONS

- Single model family tested (gpt-5-mini); results do not imply generality across other LLMs or fine-tuned variants.
- Task corpus skew: synthetic/consented tasks yielded near-ceiling baseline success ($\approx 99.75\%$), limiting headroom and statistical power to detect practical improvements by guarded pipelines.
- Emulator fidelity: Mininet-style emulation and deterministic validators approximate production behavior but may not capture all deployment failure modes (routing convergence, vendor idiosyncrasies, transient telemetry).
- Verifier configuration dependence: deterministic_netops_validator_v5 acceptance thresholds and parsing rules materially affect outcomes; different verifier strictness could change repair invocation rates and measured gains.
- Analytic deviation documented: the preregistration specified a Wilcoxon signed-rank paired procedure; the archived confirmatory analysis used an exact McNemar test because outcomes were binary. This deviation is recorded in the archived analysis artifacts and in the Disclosure Statement above.
- Operational external validity: absence of heterogeneous vendor renderers, real production templates, and live rollouts constrain direct production generalization.

VII. CONCLUSION

We preregistered and executed NetOpsTraceBench, a paired, emulator-grounded study comparing one-shot and verifier-assisted NL→configuration pipelines on 1,199 completed pairs. Under our synthetic task corpus, deterministic validator settings, and gpt-5-mini as requested, baseline one-shot generation already achieved near-ceiling success ($\approx 99.75\%$). The guarded pipeline increased absolute success by ≈ 0.25 percentage points (paired difference = 0.0025020851) with exact McNemar two-sided $p = 0.25$ and a 95% bootstrap CI that includes zero. Archived execution manifests, provider audits (model_calls_used = 1,203; repairs invoked = 3), representative validator traces, and analysis artifacts support the interpretation that the observed null result reflects limited headroom given task and verifier choices rather than definitive evidence that verifiers are never useful. Future work should intentionally design more challenging and adversarial task sets, vary verifier strictness, and test multiple LLM families to

determine conditions under which verifier scaffolding yields operationally meaningful improvements. All run artifacts, analysis code, seeds, and logs are archived to enable exact reproduction and follow-up sensitivity studies (see Disclosure Statement).

DISCLOSURE STATEMENT

Mandatory disclosure (preregistered run rX4f7-1). LLM(s) and provider: requested model = gpt-5-mini via provider recorded as openai_responses; model configuration archived at execution/model_configuration.json (requested_model=gpt-5-mini, max_output_tokens=1500). Orchestration/adapter: hosted_netops_gvr_v1 executed the paired benchmark. Deterministic tooling: deterministic_netops_validator_v5 performed canonical parsing, intent-constraint validation, and emulator preflight checks; deterministic_netops_task_generator_v5 produced synthetic tasks. Exact initial master prompt: archived at provenance/master_prompt.txt (immutable reference SHA-256 = 1872df1e1805d2d96940456ca016bd665d1d5196add77f5a cdf1582bb39b15ba). Topic constraint and autonomy boundary: the human Research Director selected the topic family and approved the master prompt pre-lock; after the autonomy lock the AI autonomously formulated the research question, conducted literature review, generated the preregistration, executed the experiment, performed analysis, and wrote and revised the manuscript. Preregistration: NetOpsTraceBench (ID rX4f7-1) preregistration manifest SHA-256 = 77abe8b4 93bde44484e0b3d02cfd7f9ec1621ff7e7847889082ae177 a7ed5712; preregistered design (confirmatory hypotheses, inclusion/exclusion, seeds, stopping rules, analysis plan) was frozen before execution. Execution summary: started 2026-08-30T01:21:55.415955+00:00, completed 2026-08-30T03:15:12.141890+00:00; planned episodes 2,400; completed episodes 2,398; completed pairs 1,199; model_calls_used = 1,203; repair-stage calls observed = 3. Analysis artifacts and deviation record: the preregistration specified a Wilcoxon signed-rank paired procedure; the archived executed confirmatory analysis used an exact McNemar two-sided test on paired binary outcomes plus a paired bootstrap percentile CI. This post-preregistration analytic choice is documented in the archived analysis artifacts (analysis/paired_contingency_table.csv, analysis/results.json, analysis/analysis_log.jsonl, analysis/paired_difference_figure.svg) produced as part of the run that completed at 2026-08-30T03:15:12.141890+00:00. Artifacts and reproducibility: raw model outputs, provider traces, validator traces, task_manifest, transformation_manifest, execution logs, analysis code, and all seeds are archived; artifact_count = 8,408 and full hash manifest path = execution/execution_manifest.json. Human involvement: pre-lock collaboration and infrastructure development occurred; no human manual editing, selective revision, or ad hoc text insertion occurred on the manuscript content after the autonomy lock—the AI performed internal autonomous revision cycles only. Technical restarts and

deviations: execution manifest records two failed episodes and the analytic deviation described above; all deviations are logged in execution/execution_log.jsonl. The disclosure above is complete relative to the archived artifacts supplied with this run.

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